Light as a Tool for Deep Object Form Perception — An Experimental Study

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Abstract
Through an experimental approach, this paper investigates light as a tool to achieve deep perception of three-dimensional object forms, to support aesthetic assessment during the creation of physical models. With product design practice as the foundation, my motivation builds on the fact that light strongly affects the designer's capabilities for assessing, modifying, and refining physical shape during the design process. In laboratory experiments with models representing defined form idioms, this study demonstrates how light may enable new and deep object form perception. This body of visual documentation supports an understanding of light's potential as both an analytical and creative resource, aiming at pedagogies that stimulate aesthetic sensitivity. Few research studies describe this specific topic; experience from teaching product design suggests that a broader group of creative professionals could benefit from building awareness of light's impact more broadly in the design industries.

Keywords
Light
Experiments
Perception
Object form
Aesthetic assessment

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Light as Phenomenon

Light is an omnipresent and life-giving element of our lives. From a human point of view, light defines the universe because it enables us to observe it, and light enables us to perceive and understand the appearance of our surroundings (Edensor, 2017). Reflected light caused by illumination enables us not only to perceive surfaces, but also spaces between surfaces. Light is often said to have the potential of improving functionality, orientation, and sense of clarity. It is measurable, yet intangible. It is form-defining, while still having a blurring effect. It may be perceived as static, while being the fastest moving medium in the universe (MacKay & Oldford, 2000). These somewhat intriguing properties appear contradictory, which also increases fascination for the different aspects of light as a natural phenomenon.

Light can be researched through diverse approaches, from large architectural scale, focusing on landscape and buildings, to the opposite small object-scale focusing on body-related products and artifacts (Zaikina, 2014). Light as a material and immaterial phenomenon has been investigated through a range of different approaches and disciplines, from social, cultural, psychological, physiological, biological, perceptual, emotional, artistic, to technical points of view (Palmer, Graham, Zezulka, 2024; Ditchburn, 1952). However, this paper moves away from viewing light as a phenomenon, to investigating light as a creative design tool that I argue may be more fully appreciated through the adoption of an experimental mindset that works with the material properties of visible light (NASA, 2010).

A core competence of a product designer is three-dimensional form giving, materialized through simulation tools and physical model building. To achieve this, three-dimensional visual analysis is a key design skill (Akner-Koler, 2007). Whether hand-made, 3D-printed, machined, or produced by other means, the development of a physical model requires light of a certain quality as a guiding tool for the creator to deeply understand the object form and to visually control object form properties. While frequent iterations during 3D printing processes (Killi, 2017) enable speed and flexibility, crafting methods and aesthetic assessment through object form perception nevertheless still depend on the quality of present light.

Topic

This experimental case study examines the influence of light on human perception to understand physical object form through the observation of real objects, excluding on-screen visualization. Based on experience from product design practice, my research question is: *How can light be utilized as a tool to explore form properties in human-scaled object forms and to obtain a deep object form perception?*

The study goes beyond the common understanding of three-dimensional form perception. The term *deep object form perception* (DOFP) is used to refer to a conscious state of mind stimulating the imaginative capability to fully comprehend the present properties of an object form, nurturing formal optimization. DOFP

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enables confirmation of a successful design, complying with a given design brief or confirming the need for modification and recreation of a provisional object form to achieve an optimal aesthetic solution. DOFP is an intentional act. It facilitates object form development during the cycle of *create-assess-recreate*, as an interactive and iterative process in the search for optimization through the perception of volume, structure, texture, and material expressions.

To be able to control light's multifaceted properties, this inquiry has been carried out in laboratory experiments. The insights described here have their origin in product design, but they should be relevant as general knowledge and applicable to all creative professions in which physical matter is created. To the best of my knowledge, these experiments are the first ones to use observation parameters in the manner described.

Literature

Several research fields may be drawn upon to accentuate ways that light may be understood in the workings of deep object form perception. With reference to object form perception theory, Peterson (2008) discusses the hierarchy of contour, shape, part, and object. These elements seem relevant as generic principles for the recognition of all physical matter. In the chapter *Balance* in his book *Art and Visual Perception*, Arnheim (1960) argues that "... no object is perceived as unique or isolated. Seeing something means assigning it a place in the whole: a location in space, a score on the yardstick of size or brightness or distance."

Perception of three-dimensional form also appears in the research and covers multi-modal observation parameters, such as the article *Visual and Haptic Perception of Three-Dimensional Solid Forms* (Garbin & Bernstein, 1984). In *Principles of Object Perception*, Spelke (1990) focuses on object perception in cluttered, changing settings, and thus reflects the norm in ordinary environments.

Interestingly, the quality of light is never mentioned as an observation parameter in any of these papers. Neglecting the influence of light on the perception of three-dimensional form suggests that light quality as an experiment parameter seems generally underestimated and undervalued within this research domain. Michel (1996) argues that within the theory of form perception, there is little agreement on how to use the terms *form* and *shape*. His use of the term form usefully designates the three-dimensional material objects resting or moving within the free volume of space, while the term shape is reserved for designating the peripheral contour of the object, and defining any two-dimensional configuration of a surface. Consequently, I will use the term *object form* as a general description of the illuminated objects in the current study.

Luminance is also a key term. Simply put, luminance (candela per square meter, SI- units cd/m^2) is defined as the intensity of light emitted from a surface per unit area in a given direction (Ditchburn, 1952). This physical attribute is vital to our understanding of light's properties as part of our study. In *Light Modelling in Architectural Spaces*, Zaikina (2016) discusses the phenomenon of *light modelling* applied to the smaller object scale. She argues that

the better light modelling is performed, the easier 3D objects can be distinguished from the background, enabling a better understanding of 3D shapes and form characteristics, as the contours, shapes and details become more clearly visible.

Several research institutions have been investigating how light influences our surroundings, and most of them focus on the larger architectural scale. The Light and Color Centre at the Norwegian University of Science and Technology, for example, has developed two highly advanced laboratories, mostly focusing on the larger architectural scale (Matusiak & Anter, 2013). However, few studies take up light as a tool for deep object perception.

Regarding the value of physical matter through model building, Lucci and Orlandini argue in *Product Design Models* (1990) that to be guided by the material is a way of interpreting reality while acquiring more experience. Furthermore, in *Form & Formlessness,* Akner-Koler (2007) argues that it is through interaction with tangible forms that one can learn to recognize the organizational capacity of form. Akner-Koler's work has become an essential inspiration for my study as it deeply investigates the intricacy of the universe of human scaled object form.

Humans' capacity to interpret structure from motion and via interaction with tangible forms has also received attention in studies in interaction design and kinetics. The paper *Motion Sequence Analysis in the Presence of Figural Cues* (Sinha & Vaina, 2014) relates to design research in the doctoral thesis *Communicating Movement. Full-body Movement as a Design Material for Digital Interaction* (Hansen, 2014), which positions choreography and movement in relation to design. This positioning supports our understanding of movement as information, and, by that, the idea that motion contributes to supporting the spatial understanding of three-dimensional form. Correlating with this, Akner-Koler (2007) stresses the value of using a turntable as a tool and rotation as a method for supporting an awareness of gestalt and ensuring the totality of the observation experience.

In their paper *Dynamic Visualization in Three Physical Dimensions*, Rowe and Morrison (2009) argue that "The use of fixed points of multi-colored light to stimulate motion and image is not new. Television is a good example. The brain smooths over the gap between the dots, actively creating (subjective) information where there is none." (p. 2). This argument supports my sense and understanding that a physical surface can be regarded as information. Furthermore, this idea has motivated me to consider the relation between object, light source, and the observers' eye(s) as separate spatial components, enabling an infinite number of dynamic constellations.

My quest for exploring a state of deep object form perception strengthens my assumption that knowledge about the relationship between light, object, and observer cannot be simply learned, it must also be experienced through personal observation. Such experimental aspects of designing are connected to what in *Three-Dimensional Visual Analysis*, Akner-Koler (1994) describes as form-taxonomy, by systematizing the broad spectrum of physical form into a hierarchy of form categories. By studying how light affects our perception of form, designers may simulate how products are perceived in real life conditions.

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Methods and Tools

To contextualize the following experiments with regard to the practice of product design, three different procedural stages of a creative process are simulated using the principle of form taxonomy. To demonstrate the typical stages of a design process, and to exemplify the convergent and reductionistic process that occurs before finalizing a form proposal, a set of hand-sized models were developed. They were categorized based on two main form idioms: geometric (GEO1/GEO2/GEO3), and organic (ORG1/ORG2/ORG3). Each of these two were divided into three different stages of completion. The first set, GEO1/ORG1, represents the initial form concept phase with rough-idea descriptive models, while the second set, GEO2/ORG2, represents a form development phase with form descriptive models. The third set, GEO3/ORG3, represents a product finalization phase with smooth, surface descriptive models. To avoid stray light, the experiments were conducted inside three closed compartments. Three different physical lucent devices were used, each for different purposes; a Black Box was used for artificial, spot based light sources, a White Box was used for abundant light, and a Solar Exploratorium was used for sunlight exposure. Photography was used to capture stills from the observations, and video was used to demonstrate the effects of relative movement.

The Experiments

A structured variation of constellations between the object form, light source(s) and observer has been key for the experiments. To enable this, an observation parameter matrix was developed, organizing the potential combinations from both natural sunlight and three artificial light sources (halogen / LED / laser), with geometric and organic form idioms, and model process stages (GEO1-3 + ORG1-3). In addition to describing the elements (light source(s), observer, and object form and their position in different configurations, possible light modes were charted in the matrix, demarcating light modes such as static, flashing, rotating, oscillating, moving/hand-held etc., and model modes (static, rotating), as well as relative movement. The physical components included in the experiments are described in Table I.

Tab. I
Overview of tools, light sources, and their properties.

Observation tools	Category	Light source		Wavelength visible, nm	Focus of experiments
Black Box	Non-parallel beams	Halogen		400-700	Artificial/spotlight sources
		Laser	Red	620-699	Spot/line/cloud luminance
			Blue	400-500	Spot/line/cloud luminance
White Box	Light abundance	LED, Light Emitting Diode		275-950	Cast shadow eliminated
Solar Exploratorium	Parallel beams	Natural sunlight		400-800	Tangential exposure

Discussion and Finding

Given the volume of documentation that was produced by photo and video recording, and the restrictions of paper length and its 2D format, only a few examples of findings from the observation experiences may be presented here. From my experiments, the most prominent findings — identified across the variety of different observation constellations — were achieved through six fundamentally different modes:

1 combined homogeneous and tangential exposure,
2 tangential exposure,
3 spot cloud with slow rotation,
4 broad (planar), static laser combined with slow object rotation,
5 combined illumination onto rotating object from one static light source and handheld laser light pattern, and
6 applying a moving laser line onto the rotating object.



Fig. 1
Static sunlight exposure using lattice onto ORG2, combined with slow object rotation.

Fig. 1 demonstrates the principle of combined homogenous and tangential exposure. This principle requires parallel light beams which may only be obtained from natural sunlight. Furthermore, this principle is based on light emitted through a homogenous filter which distributes the light in identical rays. This setup demonstrates how the light from a single point-based light source with parallel beams is illuminated through a lattice with a homogenous pattern. This experiment enables a state of DOFP as light seems to slowly 'deform' according to the angle between projected light and object surface.

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Fig. 2
Extreme tangential exposure applied on GEO3, by planar laser onto static model.

Fig. 2 demonstrates the principle of tangential exposure. Here, the GEO3 model is static while exposed to a hand-held planar laser line. This configuration enables a very careful examination of both volume and texture, constituting an extreme case of tangential exposure. Dependent on a observer's very steady hand, this configuration between light source, observer and object form produced a state of deep object form perception through a visually rewarding observation session.

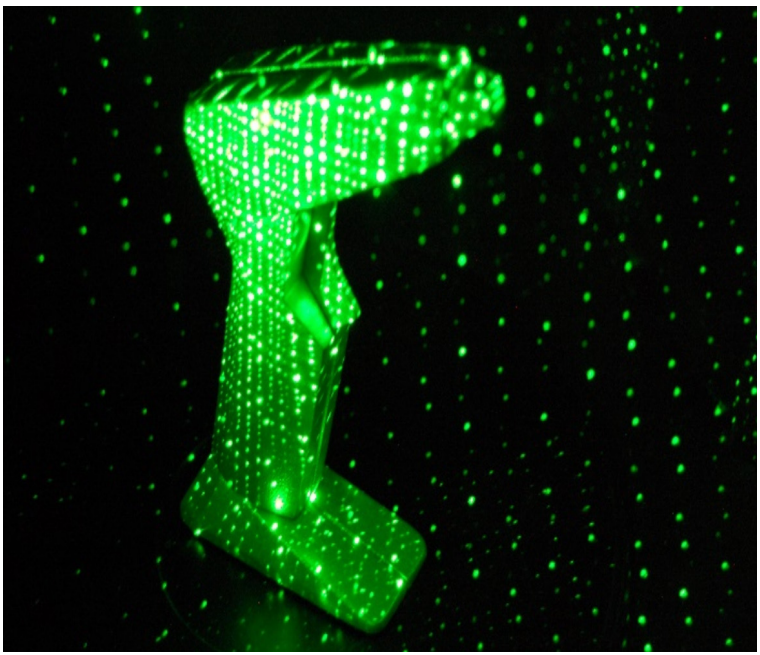


Fig. 3
GEO3 exposed to laser in spot cloud mode, combined with slow object rotation.

Fig. 3 demonstrates the principle of exposing GEO3 to a static laser in spot cloud mode combined with slow object rotation. This constellation might be regarded as far-fetched, but eventually, after some time of rotation, a strong and profound perception of volume is built through its luminousness. Interestingly, this experiment connects directly to how the brain joins the dots where information is not given, as mentioned by Rowe and Morrison (2009).

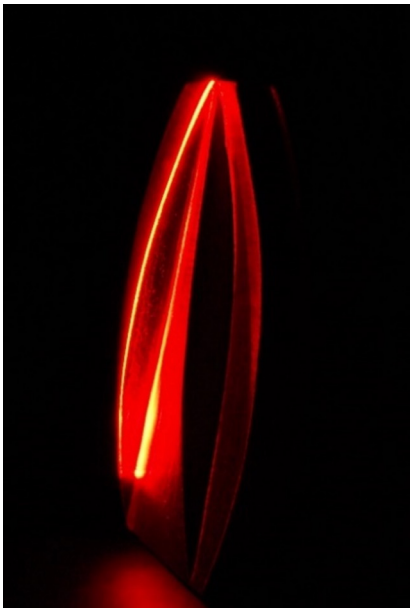


Fig. 4
ORG3 exposed to static line (planar) laser, combined with slow object rotation.

Fig. 4 demonstrates the principle of exposing ORG3 to a static, broad (planar) laser, combined with slow object rotation. This experiment reveals how an external, complex object form curvature is exposed and conveyed through a single, straight line of intense light. This might seem like an experiment with little potential outcome, but this constellation produced a clear imaginary sense of asymmetry.

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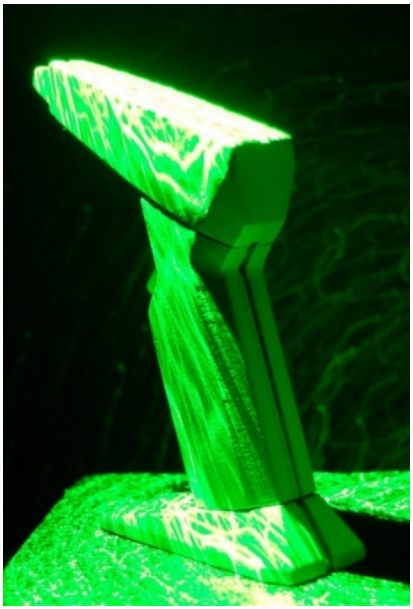


Fig. 5
GEO3 exposed to a dynamic spot-cloud laser combined with slow object rotation.

Fig. 5 demonstrates the principle of combining illumination from one static light source and laser light pattern enabled by a dynamic and random movement by hand, with slow object rotation. The laser produces evenly distributed light spots. This setup is technically complex, but the result is an intriguing experience of depth and volume, not attainable under normal light conditions. Noticeably, this experiment configuration reveals small imperfections in the model's surface, not previously observed.

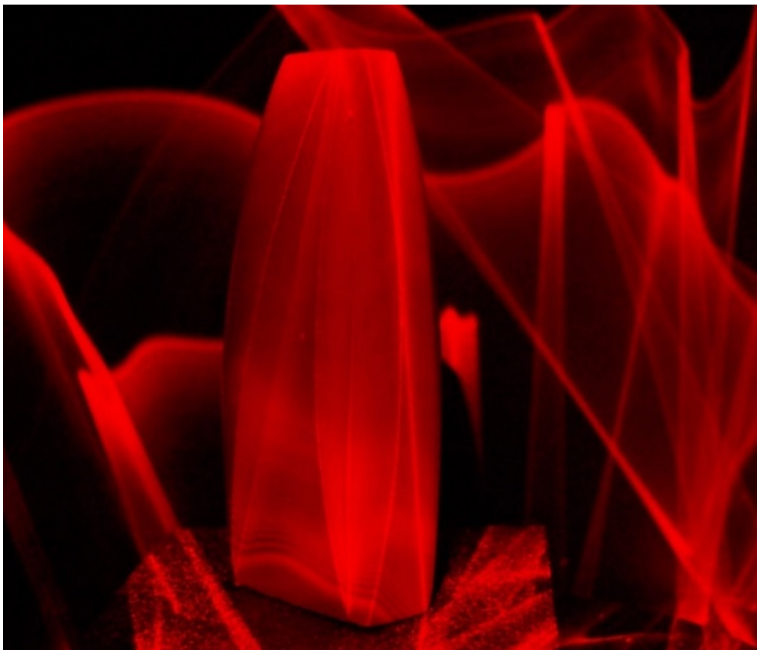


Fig. 6
Dynamic, hand-held line laser (planar), combined with slow object rotation.

Fig. 6 demonstrates the principle of applying a dynamic, moving laser line onto an object surface, combined with slow object rotation. This laser is planar, providing a straight line that enables 'painting' with light, almost like using a broad paintbrush. This approach produces a remarkably clear contour against the background, which contributes to distinguishing the object from its background, enhancing the object's proportions.

Conclusions

Light as a design material has proven to be an interesting domain of inquiry as suggested by the six key experiments assembled here. In summary, the research presented correlates with a physical-perceptual-conceptual method of reasoning which Akner-Koler (2007) describes in *Form & Formlessness* as a constructionist learning theory which "... working back and forth between concrete experiences and abstractions enhances cognitive activities." (p 57). During these experiments — and true to the nature of explorative research — both methods and tools developed progressively as the project advanced.

During several of the experiments, a state of satisfying deep object form perception was achieved. Limited to only 6 images and a written format, this paper is unable to fully support the readers' experience of the rich nature of light. However, I have demonstrated some examples of how light can be utilized as a tool to explore form properties in human-scale object form and to arrive at a sense of deep object form perception. Yet, the paradox of restricted communication and documentation adopted here is problematic. The use of photography and video recordings has no active part in the study, as deep object form perception does not happen through an optical lens due to it virtually being a sensorial filter. Nonetheless, photography and video have been necessary media and tools for the purposes of documentation. In this regard, I suggest there is value in recognizing that deep object form perception is most of all a very conscious state of mind.

In my experiments, the deep object form perception that took place required the following elements:

- Light exposure is needed either at a tangential/oblique or perpendicular angle, depending however on the light source, object form, and point of view.
- Relative movement adds to the perception of both object form, volume, and surface, enabled by rotation on a motorized turntable: an appropriate speed was found to be between 5-10 rpm.
- Mindset is key, as deep object form perception requires a conscious state of mind.
- Time is required, as both the technical set-up and observation are time-consuming activities.

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Most importantly, these experiments demonstrate the enormous complexity inherent in light as a medium. I acknowledge that the number of potential observation parameter combinations in the current work here is virtually infinite. It is impossible to complete these kinds of experiments, as one experimental set-up reveals several new alternatives to develop and test further, producing an overwhelming magnitude of potential observation combinations and constellations.

My experience from over three decades of building physical models and prototypes in both professional design practice and design education indicates that the conscious use of light enhances the clarity of form, and is crucial for precision. In the process of creating physical artefacts — from early mock-ups to highly developed prototypes — human nature may naturally cause form-related imperfections. However, during the cycle of *create-assess-recreate*, the conscious use of light through various iterations may facilitate the careful optimization of form features that could otherwise cause unwanted effects, such as unintentional light reflection, or uneven product symmetry. For organic shapes in which single curved and double curved surfaces are dominant, light's ability to reveal imperfections, especially using tangential exposure on slightly curved curvatures, is striking.

To exemplify how light may contribute to the product design process, there are two typical form-related problems that often occur in product design and are noteworthy. The first is apparent in cabinet design when enclosing internal components, a common design task in metal-based production processes, and when using high-pressure injection moulding for polymer materials. Here, an optical problem often occurs in the case when the design involves a planar surface. Due to the undesirable and uncontrollable light reflections that may occur here, the designer must try to avoid a perfectly planar surface and instead shape the visual plane into a slightly double-curved surface to control the light reflected from the surface. The glossier the material surface, the more important this measure is. The second example is found in body design within the automobile industry, where the design prototypes are moved slowly through light tunnels equipped with linear light sources in parallel configurations, enabling an extremely careful examination of light reflections to discover imperfections in the car bodywork. Both cases explain design considerations in which the appearance of light might be examined through rotation on screen in a CAD program, producing *informed* light effects. However, only a physical, real-world examination using actual light will reveal the intended light reflection on the surface of the final product design, resulting in *confirmed* light effects.

To argue the research results, especially in relation to the field of design, it seems appropriate to take into consideration the various aspects of the product, and not only the formal ones. Consideration of the practical implications of this study on today's design practice leads us to take a multi-faceted view of the implications of utilizing light as a creative resource in relation to product design. It is of fundamental importance for a product designer to predict how a product will appear in its natural environment based on its intended use, and experimenting with light becomes a catalyst to make this prediction.

To report results in a more didactic way and allow the reader to consider them more directly, the following topics describe potentially relevant aspects:

- Form expression. Light may act as a guide when creating form, especially for products with semi-organic bodies and other complex shapes, such as advanced transport vehicles and sport and leisure products. This ability seems valuable within many product categories.
- Design teaching. Pedagogy in product design education can intentionally utilize light as a creative tool. Insights and reflections based on light-object observations may inform classes and exercises that teach object form assessment. These insights may strengthen design students' ability to develop form consciousness and designerly skills, and their ability to argue for aesthetic advantages as well as semantic values inherent in a design solution.
- Design profession. Light affects the designer's capabilities for assessing, modifying, and refining physical shape during the design process, and this may enable untrained designers to excel.
- Communication tool. Just as light enables a creator to obtain clarity of form, a professionally developed product design with clear features may communicate strongly to a given audience.
- Industrial applications. As described in the example of using light as a correction tool within the automobile industry, light can be utilized for quality assurance in manufacturing on a large industrial scale. There are also examples of how light is used to determine quality in food production, where the quality observed from object contours distinguishes accepted items from unaccepted items. When taken further, light has unlimited potential for use as a correction tool in many production applications, including robotics and automation.
- Consumer appreciation. In providing better solutions for the consumer in the form of more aesthetically appealing products, designers possess a powerful tool to develop attractive objects to approach mass-markets with both consumer goods as well as capital goods.
- Artistic purposes. As seen amongst several contemporary artists and designers, exploration with light may create optical illusions or distort the perception of physical reality using light, but in that case a more intentionally artistic approach is taken.

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To clarify the objectives, contexts, and methods of the relationship between light and the perception of form, it is important to emphasize that my approach deviates from other approaches to assessing object form properties, as those approaches do not include light as a parameter. The common practice in design teaching in higher education when reviewing a physical model is to assess the physical model or prototype under general light conditions without any focus on the quality of the actual light. I have not come across similar studies in which light contributes to examining the influence of light on human perception, to understanding physical object form through the observation of real objects.

Future Work

Despite its intangibility and immaterial characteristics, light has several properties that are comparable to physical matter. Light is measurable, scalable, positionable, movable, and modular. These properties need attention to further support our understanding of light as a highly operational tool. Such aspects would be interesting to explain and describe more deeply in future research. Furthermore, it would be an interesting challenge to convey the observation parameter matrix presented here in a more complete form that describes and presents all the parameters.

In closing, I argue that we lose something valuable when we are unable to convey more fully and three-dimensionally the satisfying and rewarding state of deep object perception of light in this paper format. However, in further work it could be interesting to figure out how these sensorial, haptic, and spatial experiences might be conveyed to a broader audience and in different communicative formats.

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